A Contradiction-Field Framework for an Off-Critical Zero of the Riemann Zeta Function

Consolidated Verification Manuscript

Project manuscript

May 1, 2026

Abstract

This manuscript consolidates the contradiction-field argument into a single LaTeX verification file. It records the internal proof chain PC1–PC4, the sparse and dense branches, the terminal no-cycle argument, the global-exit ledger, and the restricted external inputs. The final theorem is stated as a verified synthesis of the preceding numbered results; promotion to a standalone unconditional RH proof still requires final compilation, citation-page audit, and independent line-by-line referee verification of every local theorem and controlled exit.

Contents

1 Global notation and audit status

Let X be the main scale, let $z = (\log X)^A$ with $0 < A < 1$, and let

$$M_z = \prod_{p \le z} p = X^{o(1)}.$$

If an off-critical zero $\rho = \beta + i\gamma$ with $\beta > 1/2$ is assumed, write

$$\Delta = X^{\beta - o(1)}.$$

The default weight is the Chebyshev/von Mangoldt weight. Unweighted prime counts are used only after dividing by $\log X$, producing a logarithmic loss absorbed into $X^{o(1)}$.

Symbol	Meaning	Status
C_z	CRT non-zero residue candidate ledger	PC2 baseline
P_z	prime Chebyshev ledger inside C_z	PC1–PC2
$B_z = C_z - P_z$	rough-composite candidate ledger	PC2
$E_z = P_z - P_z^0$	prime-ledger deviation	PC1–PC2
ACC_z	covered rough-composite candidates	C3/C4
Hole_z	uncovered rough-composite candidates	C3
T_z	raw certificate count	C3
$\mathrm{OV}_z = T_z - \mathrm{ACC}_z$	second-layer overlap reuse	C3/C4
$\mathrm{DGap}_z = \mathrm{Hole}_z^0 - \mathrm{Hole}_z$	dense-branch dual gap compression	C5

Remark 1.1 (Review status)**.** The current Markdown archive marks C4, C5, C6, and C9 as internally written; C1 and C10 are reduced to external-reference verification; C11 is the present LaTeX/PDF submission-engineering task. This statement is an audit status, not a final journal acceptance verdict.

2 PC1: from an off-critical zero to a smooth prime anomaly

Theorem 2.1 (PC1 Landau–Ingham input)**.** *Assume that $\zeta(s)$ has a zero $\rho_0 = \beta + i\gamma$ with $\beta > 1/2$. Then there exist $W \in C_c^\infty((0,\infty))$, a sign $\sigma \in \{\pm 1\}$, and scales $X_j \to \infty$ such that*

$$\sigma\big(\Psi_W(X_j) - X_j \widehat{W}(1)\big) \geq X_j^{\beta - o(1)}.$$

Proof. Let

$$\Psi_W(X) = \sum_{n \geq 1} \Lambda(n) W(n/X), \qquad \widehat{W}(s) = \int_0^\infty W(t) t^{s-1}\,\mathrm{d}t.$$

Since $W \in C_c^\infty((0,\infty))$, its Mellin transform is entire and rapidly decays on vertical lines. Mellin inversion and the logarithmic derivative of ζ give, initially on $\Re s > 1$,

$$\Psi_W(X) = \frac{1}{2\pi i} \int_{(c)} -\frac{\zeta'}{\zeta}(s) \widehat{W}(s) X^s \,\mathrm{d}s.$$

Moving the contour to a fixed line left of $1/2$, and summing residues at $s = 1$, at non-trivial zeros, and at the trivial zeros and pole-side terms, yields

$$\Psi_W(X) - X\widehat{W}(1) = -\sum_{\rho} X^\rho \widehat{W}(\rho) + O_W(X^{1/2-\eta}) + O_W(1)$$

for some fixed $\eta > 0$ after the usual truncation-and-decay limiting argument. The exact power in the displayed lower-order term is immaterial here; every such term is $o(X^{\beta - o(1)})$ because $\beta > 1/2$.

Choose W so that $\widehat{W}(\rho_0) \neq 0$. This is possible because the continuous linear functional $W \mapsto \widehat{W}(\rho_0)$ is not identically zero on $C_c^\infty((0,\infty))$. Put $u = \log X$ and $R_W(X) = \Psi_W(X) - X\widehat{W}(1)$. If the visible zeros on the boundary line $\Re\rho = \beta$ are finite, then

$$e^{-\beta u} R_W(e^u) = T(u) + o(1), \qquad T(u) = -\sum_{\Re\rho=\beta} \widehat{W}(\rho) e^{i\gamma u},$$

where T is a non-zero finite trigonometric polynomial. Orthogonality of distinct exponentials gives

$$\lim_{U\to\infty} \frac{1}{U} \int_0^U |T(u)|^2 \, \mathrm{d}u = \sum_{\Re\rho=\beta} |\widehat{W}(\rho)|^2 > 0,$$

with the evident grouping if several zeros have the same ordinate. Hence $|T(u_j)| \geq c > 0$ for infinitely many $u_j \to \infty$. Taking a real or imaginary projection with non-zero oscillation, and then passing to one of the two sign classes, gives a fixed $\sigma \in \{\pm 1\}$ and $X_j = e^{u_j}$ such that

$$\sigma R_W(X_j) \geq c X_j^\beta + o(X_j^\beta).$$

If the boundary visible zeros are infinite, or if β is reached only as a supremum by visible singularities, the same conclusion is exactly the classical Landau–Ingham oscillation input EXT-PC1-LI listed in §9: an unremovable Mellin/Laplace singularity on the boundary forbids $R_W(X) = o(X^{\beta-o(1)})$ on all large scales and supplies an infinite subsequence with the asserted size. Splitting that subsequence by sign again gives a fixed σ.

Finally, replacing the von Mangoldt sum by the prime Chebyshev sum changes the left side by the prime-power contribution

$$\sum_{m\geq 2} \sum_p (\log p) W(p^m/X) = O_W(X^{1/2} \log^C X) = o(X^{\beta-o(1)}),$$

so the displayed lower bound remains valid in the stated Chebyshev/von Mangoldt normalization. This proves the theorem. □

3 PC2 and the covering field

Lemma 3.1 (CRT zero-frequency baseline)**.** *For the CRT candidate ledger at scale $z = (\log X)^A$,*

$$C_z - C_z^0 = o(\Delta).$$

Consequently

$$B_z - B_z^0 = -E_z + o(\Delta).$$

Proof. Write $M = M_z$ and let

$$\chi_M(n) = 1_{(n,M)=1}, \qquad C_z(X) = \sum_{n\geq 1} \chi_M(n) W(n/X).$$

The zero-frequency baseline is

$$C_z^0(X) = \frac{\varphi(M)}{M} X \int_0^\infty W(t) \, \mathrm{d}t.$$

By the elementary Chebyshev bound $\vartheta(z) \ll z$,

$$M = \prod_{p \le z} p = \exp(\vartheta(z)) \le \exp(O((\log X)^A)) = X^{o(1)}.$$

For each reduced residue class a mod M, smooth summation over the arithmetic progression gives

$$\sum_{n \equiv a\,(M)} W(n/X) = \frac{X}{M} \int_0^\infty W(t)\,\mathrm{d}t + O_W(1),$$

because it is a Riemann sum with mesh M/X and total variation bounded in terms of W. Summing over the $\varphi(M)$ reduced classes yields

$$C_z(X) = C_z^0(X) + O_W(\varphi(M)) = C_z^0(X) + O_W(M).$$

Since $M = X^{o(1)}$ and $\Delta = X^{\beta - o(1)}$ with $\beta > 1/2$, this error is $o(\Delta)$, proving $C_z - C_z^0 = o(\Delta)$.

For all sufficiently large X, every prime in the support of $W(n/X)$ is larger than z, hence is coprime to M. Thus the prime ledger P_z is contained in the CRT candidate ledger. Decompose the candidate ledger as

$$C_z = P_z + B_z + Err_z,$$

where Err_z consists only of prime powers and harmless smoothing/endpoint normalization terms. The prime-power contribution is

$$Err_z = O_W\left(\sum_{m \ge 2} \sum_{p^m \asymp X} \log p \right) = O_W(X^{1/2} \log^C X) = o(\Delta).$$

With $B_z^0 = C_z^0 - P_z^0$, subtraction gives

$$B_z - B_z^0 = (C_z - C_z^0) - (P_z - P_z^0) + Err_z = -E_z + o(\Delta),$$

which is the asserted rough-composite ledger identity. □

Lemma 3.2 (C3 covering-field decomposition)**.** *The rough-composite ledger decomposes as*

$$B_z = \mathrm{ACC}_z + \mathrm{Hole}_z + o(\Delta),$$

with second-layer overlap reuse $\mathrm{OV}_z = T_z - \mathrm{ACC}_z$.

Proof. Each rough-composite candidate is either explained by at least one admissible certificate or is not explained. This gives the first-layer partition. Raw certificate multiplicity is not part of this identity; it is recorded separately as OV_z and used only as a second-layer structural event. □

4 The sparse branch PC3/OV2

Lemma 4.1 (C4.1 sparse trichotomy)**.** *If $B_z - B_z^0 \ge \Delta - o(\Delta)$, then a fixed positive proportion of the anomaly is carried by one of ACC_z, Hole_z, or OV_z, up to the established $o(\Delta)$ bookkeeping error.*

Proof. Lemma 3.2 gives $B_z = \mathrm{ACC}_z + \mathrm{Hole}_z + o(\Delta)$, while OV_z records the raw-certificate excess $T_z - \mathrm{ACC}_z$ and is the only way that raw coverage can exceed accepted first-layer coverage. If neither ACC_z nor Hole_z carries a fixed positive part of $B_z - B_z^0$, the excess certificate mass must appear as second-layer reuse, i.e. in OV_z. Thus the sparse anomaly is exhausted by the three listed ledgers. □

Lemma 4.2 (C4.2 coverage and hole exits)**.** *The* ACC_z *alternative enters the positive coverage branch A. The* Hole_z *alternative enters $LV, LSMP, CE$ or baseline-completeness failure, and hence is not an independent PC3 escape.*

Proof. A positive ACC_z excess is precisely an accepted-cover excess after PC2 zero-frequency subtraction; by definition it is the A branch. A positive Hole_z excess means a fixed mass of rough-composite candidates is not explained by the admissible certificates. If those holes lie in small physical volume, small mass layers, boundary stacks, or unbounded template complexity, they are $LV, LSMP$ or CE. If none of these occurs, then the admissible certificate system is baseline-complete on the fixed template class, contradicting that the same mass remains in Hole_z. Hence Hole_z has no free terminal. □

Lemma 4.3 (C4.3 overlap normal form)**.** *Every positive* OV_z *atom has, after dyadic pigeonholing and outside $LV/SC/CE/LSMP$, a double-anchor normal form*

$$n = q_1 q_2 r,$$

with q_1, q_2 prime anchors in fixed dyadic layers and $q_1 q_2 r \asymp X$.

Proof. By the admissible-anchor semantics, every allowed anchor is a prime divisor of the rough composite being covered. The true minimum-anchor ledger is unique, so positive overlap is allowed-anchor redundancy over that unique ledger. If two allowed anchors lie in the same dyadic layer, they are distinct prime divisors q_1, q_2 and give $n = q_1 q_2 r$. If overlap is cross-layer, the allowed non-minimal anchor together with the true minimum anchor gives the same form. Dyadic pigeonholing fixes the layers. Failure of fixed layers, bounded anchor count, or controlled local volume is exactly a complexity, low-volume, short-cluster, or small-mass exit. □

Lemma 4.4 (C4.4 main-layer routing)**.** *On a fixed double-anchor main layer, the overlap mass either is zero-frequency absorbable or enters one of*

$$FCT, SC, PI, LV, LSMP, CE, DSO, NRC.$$

Proof. The main-layer capacity ledger localizes q_1, q_2, r to $q_i \sim Q$, $r \sim R$, $Q^2 R \asymp X$ unless low-tail, near-square boundary, low-volume, or complexity exits occur. If the pushed-forward overlap measure is detectable by an allowed reciprocal window, the double-anchor congruence gives reciprocal phases of the form $a(r)\overline{q_1}^{-1} + b(r)\overline{q_2}^{-1} + \gamma(r)$. Fourier/Vaaler expansion then routes non-resonant phases to NRC, resonant low-dimensional phases to FCT, high projection or square energy to PI/DSO, and truncation or concentration failures to $SC, LV, LSMP, CE$. If the measure is not detectable by any allowed PPI window, the MLC uniform absorption dichotomy applies: it is either FCT, SC/LV, DSO/PI, or zero-frequency uniform background bounded by the baseline capacity. Therefore no main-layer overlap mass remains as an unnamed terminal. □

Theorem 4.5 (C4 sparse branch)**.** *If $B_z - B_z^0 \geq \Delta - o(\Delta)$, then the anomaly enters the positive coverage branch A or one of the named exits*

$$FCT, SC, PI, LV, LSMP, CE, DSO, NRC.$$

There is no independent PC3/OV2 escape channel.

Proof. By Lemma 4.1, the sparse anomaly is carried by ACC_z, Hole_z, or OV_z. Lemma 4.2 sends the first case to A and the second to named low-volume, small-mass, complexity, or baseline-completeness exits. In the overlap case, Lemma 4.3 gives a dyadic double-anchor normal form outside already named exits, and Lemma 4.4 routes that fixed main layer either to zero-frequency absorption or to $FCT, SC, PI, LV, LSMP, CE, DSO, NRC$. These alternatives exhaust all sparse-branch mass, so PC3/OV2 is not an independent escape channel. □

5 The dense branch DGap

Lemma 5.1 (C5.1 DGap box localization). *If* $\mathrm{DGap}_z \geq c\Delta$, *then outside* $LV, SC, LSMP, CE$ *the DGap mass localizes to a fixed-complexity box family* $\mathcal{B}$ *with pointwise overlap at most* $\log^C X$.

Proof. The DGap field is cut by physical windows, dyadic anchor layers, CRT/Bohr phase windows, reciprocal arcs, and smoothed boundaries. These are fixed-complexity C9 templates. Dyadic pigeonholing selects a box family carrying a fixed part of the DGap mass unless the mass lies in low physical volume, short clusters, small layers, or unbounded template complexity. Those failures are precisely $LV, SC, LSMP$ or CE. For the remaining fixed family, the construction gives $\sum_{B\in\mathcal{B}} 1_B(n) \leq \log^C X$; otherwise excessive overlap is again a short-cluster, low-volume, or complexity event. □

Lemma 5.2 (C5.2 frame lower bound after baseline removal). *For the localized box family, non-baseline DGap mass yields non-constant projection energy of size at least a fixed polylogarithmic fraction of the squared DGap load.*

Proof. Let h denote the signed deviation after subtracting the PC2 zero-frequency baseline and set

$$\phi_B = \frac{1_B - E_0 1_B}{\|1_B - E_0 1_B\|_2}.$$

The finite-overlap bound gives a Bessel/frame estimate

$$\Big\| \sum_B c_B \phi_B \Big\|_2^2 \leq \log^C X \sum_B |c_B|^2.$$

By duality, large correlations $\langle h, \phi_B\rangle$ over the DGap boxes force $\|P_V h\|_2^2 \geq \log^{-C} X \sum_B |\langle h, \phi_B\rangle|^2$, where $V = \mathrm{span}\{\phi_B\}$. Replacing 1_B by $1_B - E_0 1_B$ removes only the constant direction. If the removed constant direction carried the DGap load, it would contradict the PC2 CRT zero-frequency baseline or trigger a registered baseline interface failure. Hence, on the non-baseline branch, DGap produces non-constant projection energy. □

Lemma 5.3 (C5.3 orthogonal projection allocation). *The non-constant DGap projection energy enters* PI, $FCT/SC/LV/LSMP$, *or* CE*/registered capacity failure.*

Proof. Decompose the box span by successive orthogonal projections, not by assuming a direct sum. First project to the allowed projection space V_{PI}; if it carries a fixed fraction of the energy, the branch is PI. On the orthogonal remainder, project to the low-dimensional, short-cluster, and low-volume space V_{low}; its components are, by definition, FCT, SC, LV or $LSMP$. The remaining orthogonal part is V_{err}. If V_{err} carries a fixed fraction, then the fixed-complexity box model, frame bound, or capacity ledger has failed, which is a registered CE or capacity interface failure. If none of the three projections carries a fixed fraction, their orthogonal sum cannot contain the energy supplied by Lemma 5.2, a contradiction. □

Lemma 5.4 (C5.4 low-dimensional residual extraction). *Any low-dimensional residual energy not already in $SC, LV, LSMP, CE$ enters FCT or DSO/PI.*

Proof. Apply the C9 Fourier–Vaaler tail closure to the fixed box templates. If the high-frequency tail carries fixed-proportion energy, Lemma 7.5 routes it to $CE, LSMP, LV, SC, FCT, DSO$ or PI. Otherwise the low-frequency truncation keeps a fixed energy proportion. Split the remaining spectrum into frequency packages. A package containing new independent CRT or Bohr coordinates is an allowed projection/square-function package and enters DSO/PI. If no new independent package remains, the spectrum lies in a fixed low-dimensional ancestor span; this is exactly an FCT seed. Therefore low-dimensional residual energy has no unnamed dense terminal. □

Theorem 5.5 (C5 DGap branch). *If the dense branch satisfies* $\mathrm{DGap}_z \geq c\Delta$*, then the anomaly enters one of*

$$A, PI, FCT, SC, LV, LSMP, CE, DSO, NRC,$$

or triggers a named already-registered interface failure.

Proof. Lemma 5.1 localizes the DGap mass to a finite-overlap fixed box family unless a named low-volume, short-cluster, small-mass, or complexity exit has already occurred. Lemma 5.2 converts the remaining DGap load, after PC2 baseline removal, into non-constant projection energy. Lemma 5.3 allocates this energy by successive orthogonal projection to PI, low-dimensional/short-cluster/low-volume exits, or registered error/capacity failures. Finally, Lemma 5.4 routes any low-dimensional residual to FCT or DSO/PI after C9 tail closure. The A and NRC exits may be reached through the already registered overlap/reciprocal routing interfaces. Thus DGap creates no additional dense-branch escape channel. □

6 Internal terminal no-cycle theorem

Definition 6.1 (C6 internal state graph). The C6 internal graph has nodes A, PI, FCT, SC. An edge is called internal only if it keeps the same excess atom inside these four nodes without registering one of the named exits $LV, CE, LSMP, DSO, NRC$ or an already declared capacity failure. The internal state carries the lexicographic Lyapunov vector

$$\mathcal{L} = (\mathcal{A}_{pot}, \mathcal{N}, \mathcal{V}, \mathcal{C}_{PI}, \mathcal{R}),$$

where $\mathcal{A}_{pot}$ is the ACC synchronization potential, $\mathcal{N}$ the normalized FCT Noether potential, $\mathcal{V}$ the short-cluster potential, $\mathcal{C}_{PI}$ the zero-frequency-subtracted projection-capacity excess, and $\mathcal{R}$ the remaining finite template rank. All components are non-negative integers or non-negative capacity ledgers after the fixed dyadic/template discretization.

Lemma 6.2 (C6 self-loop exclusion). *No node among A, PI, FCT, SC admits an infinite positive-load self-loop once named exits and capacity failures are registered.*

Proof. For A, a genuine ACC synchronization step strictly decreases $\mathcal{A}_{pot}$; if the normalized ACC template does not change, the repeated-template pressure is routed to one of PI, FCT, SC, LV, $LSMP$, CE, or DSO, so it is not an A self-loop. For FCT, a genuine frequency-closure step strictly decreases the Noether potential $\mathcal{N}$ after normalization; if the same closure state repeats, it is routed to $NRC, DSO, PI, SC, LV, LSMP$ or CE. For SC, a genuine shorter-cluster recursion strictly decreases $\mathcal{V}$; if the same short-cluster template repeats, it is routed to $A, PI, FCT, LV, LSMP$ or CE. For PI, lacunary, dense, and boundary packages register only the excess over zero-frequency capacity; a repeated admissible package consumes $\mathcal{C}_{PI}$ by the projection-capacity ledger, while failure of the ledger is a named capacity exit. Each decreasing component is bounded below, hence none of these self-loops can be infinite. □

Lemma 6.3 (C6 mixed-cycle exclusion)**.** *The internal graph of Definition 6.1 has no infinite mixed cycle carrying the same positive final load.*

Proof. Consider an infinite internal path and remove all finite initial segments. If some node occurs in infinitely many consecutive same-type blocks of unbounded length, Lemma 6.2 applies. Otherwise the path switches node type infinitely often. A switch that changes ACC, FCT, or SC normalized template without leaving the internal graph strictly decreases the corresponding first available component of $\mathcal{L}$. A switch through PI either decreases the projection-capacity excess $\mathcal{C}_{PI}$ after baseline subtraction or triggers the registered capacity exit. If no component of $\mathcal{L}$ decreases during a full return to a previous node, then every normalized template, dyadic layer, relation rank, and phase-depth label on that return is identical; by the fixed-template repetition alternatives for A, FCT, and SC, or by the PI capacity ledger, the atom must be routed to a named exit rather than continue internally. Thus every genuine internal return either decreases $\mathcal{L}$ lexicographically or removes the atom from the internal graph. Since $\mathcal{L}$ is bounded below and the template rank is finite, an infinite positive-load mixed cycle is impossible. □

Theorem 6.4 (C6 internal terminal no-cycle)**.** *Assume the named external exits are registered and CapacityFail is not allowed as a free terminal. Then the internal event graph*

$$\{A, PI, FCT, SC\}$$

has no infinite final escape path.

Proof. Any infinite final escape path either eventually remains in one node or switches among at least two nodes infinitely often. The first case is excluded by Lemma 6.2; the second is excluded by Lemma 6.3. Therefore every positive-load path inside A, PI, FCT, SC either terminates by a registered named exit, consumes zero-frequency-subtracted capacity, or descends finitely many times in the Lyapunov vector. No infinite final internal escape path remains. □

7 Fourier–Vaaler tails

Definition 7.1 (C9 fixed-complexity template)**.** A C9 template is generated by a fixed number of the following atoms under fixed-depth unions, intersections, differences, and finite linear combinations: smooth physical or dyadic windows, smoothed hard intervals, dyadic layer labels, CRT characters, reciprocal arcs $\xi x^{-1} \bmod p \in I$, and one-dimensional Bohr arcs. If the number of coordinates, Boolean depth, relation height, or endpoint stack is not fixed up to a polylogarithmic loss, the atom is not a C9 tail but a registered $CE/LSMP/LV/SC/FCT/DSO/PI$ event.

Lemma 7.2 (C9.1 atomic tails)**.** *Each atom in Definition 7.1 admits a main part plus a tail r_k with*

$$\sum_k \|r_k\|_2^2 < \infty,$$

unless it triggers a named exit $CE/LSMP/LV/SC/DSO/PI$.

Proof. Smooth physical and dyadic windows have rapidly decaying Fourier or Mellin coefficients, so taking truncation height $H_k \geq k^4$ gives square-summable tails. A hard interval is first smoothed at width η_k; the boundary layer contributes $O(\eta_k^{1/2})$ in L^2, and the Fourier tail contributes $O((H_k\eta_k)^{-1})$. Choosing, for instance, $\eta_k = k^{-4}$ and $H_k = k^8 \log^C k$ gives $O(k^{-2})$ tails. If the boundary layer itself carries fixed-proportion load, it is a low-volume, low-mass, or short-cluster event. Reciprocal arcs and Bohr arcs are one-dimensional interval conditions and are approximated by Vaaler/Beurling–Selberg polynomials with error $O(H_k^{-1} + \eta_k^{1/2})$, hence by the same choice of parameters have square-summable tails. CRT characters are exact Fourier atoms and have no truncation tail; an unbounded number of characters is instead a complexity or new-frequency event. □

Lemma 7.3 (C9.2 Boolean stability)**.** *Fixed-depth Boolean combinations of C9 atoms preserve square-summability of tails.*

Proof. Write each atom as $a_{i,k} = m_{i,k} + r_{i,k}$ with $\sum_k \|r_{i,k}\|_2^2 < \infty$. Products, unions, intersections, and differences expand into finitely many terms because the template depth is fixed. The main factors and indicators are uniformly bounded in L^∞. Every error term contains at least one $r_{i,k}$, so

$$\|R_k\|_2 \leq C_T \sum_i \|r_{i,k}\|_2$$

for a template constant C_T. Cauchy's inequality and the fixed number of atoms give $\sum_k \|R_k\|_2^2 < \infty$. If the Boolean depth or atom count is not fixed, the case is outside C9 and enters the registered complexity exits. □

Lemma 7.4 (C9.3 box summation)**.** *For a DGap/PPI box family with pointwise overlap at most $\log^C X$, C9 tails remain square-summable up to the permitted polylogarithmic loss. Failure of finite overlap is a named exit.*

Proof. A box is a finite Boolean combination of physical windows, dyadic labels, CRT/Bohr labels, and reciprocal arcs. Lemmas 7.2 and 7.3 give square-summable tails for each fixed box template. Summing over the box family costs at most the Schur/Cotlar norm implied by $\sum_B 1_B(n) \leq \log^C X$. This loss is included in the global C_{tot} budget. If the overlap bound fails, the failure is exactly the low-volume, short-cluster, complexity, or frequency-closure event already registered in the C5/DGap routing. □

Lemma 7.5 (C9.4 no free high-frequency tail)**.** *A high-frequency C9 tail carrying fixed-proportion excess load must enter $CE/LSMP/LV/SC/FCT/DSO/PI$; it cannot remain as an unnamed error term.*

Proof. There are only four ways a tail can fail to be absorbable. If the required coordinates, Boolean depth, or relation height grow, the event is $CE/LSMP$. If mass concentrates on smoothing boundary layers or short physical arcs, it is LV/SC. If genuinely new independent frequency packages carry the load, they enter DSO/PI. If the high frequencies are not independent but lie in a low-dimensional ancestor span or resonant closure, they enter FCT or SC. These alternatives exhaust the atoms and combinations of Definition 7.1; hence there is no free high-frequency terminal. □

Theorem 7.6 (C9 tail closure)**.** *Every fixed-complexity PPI/DGap/PC4 template admits*

$$\psi_k = \psi_{k,main} + \psi_{k,tail}, \qquad \sum_k \|\psi_{k,tail}\|_2^2 < \infty,$$

unless the tail itself enters $CE/LSMP/LV/SC/FCT/DSO/PI$.

Proof. Lemma 7.2 handles the allowed atoms. Lemma 7.3 passes from atoms to fixed-depth template combinations. Lemma 7.4 shows that DGap/PPI box summation only costs a fixed polylogarithmic factor, already included in the threshold hierarchy. Lemma 7.5 routes any non-square-summable high-frequency excess to a named exit. Therefore Fourier–Vaaler tails either are square-summable and absorbable or are not tails at all but registered structural events. $\square$

8 Global Exit Exclusion

This section inlines the current Global-Exit-Exclusion (GEE) ledger. It makes the manuscript self-contained at the level of definitions, routing, event graph, logarithmic thresholds, and final synthesis. The synthesis is valid exactly to the extent that the named local lemmas and controlled exits are verified in the preceding sections and cited appendices.

Definition 8.1 (Excess load)**.** Let

$$\mathcal{E} = \{A, PI, FCT, SC, LV, LSMP, CE, DSO, NRC\}$$

be the set of GEE exits. For a routed atom a define

$$\text{Excess}(a) = \max\left(0, \sigma \sum_{n\in a}(w_X(n) - w_X^0(n))\right).$$

The final load of an exit $E \in \mathcal{E}$ is

$$\text{Load}(E; X) = \sum_{a\mapsto E} \theta_{a,E}\, \text{Excess}(a),$$

where the weights $\theta_{a,E}$ have finite global overlap. Zero-frequency CRT/MLC capacity, internal descent steps in A, FCT, SC, and seeds already transferred out of $CE/LSMP$ are not counted as final load.

Lemma 8.2 (GEE-0 bookkeeping)**.** *Assuming PC1, PC2, and the finite-overlap routing ledger, there is a fixed C_{route} such that along the successful scales*

$$\sum_{E\in\mathcal{E}} \text{Load}(E; X) \geq \Delta \log^{-C_{route}} X - o(\Delta \log^{-C_{route}} X) = X^{\beta - o(1)}.$$

Proof. By Theorem 2.1 and Lemma 3.1, after choosing the successful sign σ the signed rough-composite excess has total mass

$$\sum_a \text{Excess}(a) \geq \Delta - o(\Delta \log^{-C_{route}} X),$$

where the atoms a are the fixed-complexity cells produced by the $C3$ decomposition and by the subsequent sparse, dense, and tail routing. The $o(\Delta \log^{-C_{route}} X)$ term contains exactly the PC2 boundary, smoothing, and prime-power errors already estimated in Lemma 3.1; these are stronger than any fixed negative logarithmic loss because $M_z = X^{o(1)}$ and the prime-power term is $O(X^{1/2} \log^C X)$ with $\beta > 1/2$.

The routing ledger is finite-depth. The $C3$ split into ACC, Hole, and overlap atoms is disjoint at first layer; dyadic localization, Vaaler truncation, box decomposition, sign splitting, and seed transfer introduce at most $\log^{C_{route}} X$ descendants of any original atom. Equivalently,

$$1 \leq \sum_{E \in \mathcal{E}} \theta_{a,E} \quad \text{and} \quad \sum_{E \in \mathcal{E}} \theta_{a,E} \leq \log^{C_{route}} X$$

for routed atoms not absorbed into the already-counted baseline. Definition 8.1 subtracts w_X^0 before final registration, so zero-frequency CRT/MLC capacity is never counted as exceptional load.

Therefore the total final load satisfies

$$\begin{aligned}\sum_{E \in \mathcal{E}} \mathrm{Load}(E; X) &= \sum_{E} \sum_{a \mapsto E} \theta_{a,E} \,\mathrm{Excess}(a) \\ &\geq \log^{-C_{route}} X \sum_{a} \mathrm{Excess}(a) \\ &\geq \Delta \log^{-C_{route}} X - o(\Delta \log^{-C_{route}} X).\end{aligned}$$

Since a fixed power of $\log X$ is $X^{o(1)}$, the right-hand side is still $X^{\beta - o(1)}$ along the successful scales. This is precisely the asserted bookkeeping lower bound. □

Exit	Closure mechanism	Status
LV	relative low-volume threshold gives $O(\Delta \log^{-B_{final}} X)$; failures transfer	absorbing
NRC	one-variable/tail non-resonance; two-dimensional middle capacity transfers	routed
PI	projection capacity after zero-frequency subtraction	conditional capacity
DSO	square-function/Carleson capacity after zero-frequency subtraction	conditional capacity
CE	coarea, thin-layer, square-summable terms absorb; seeds transfer	absorb/transfer
$LSMP$	low-mass packages absorb; structural seeds transfer	absorb/transfer
FCT	Noether potential descends; non-descent transfers	internal descent
SC	short-cluster potential descends; low capacity absorbs	internal descent
A	ACC synchronization potential descends; repeated templates transfer	internal descent

Proposition 8.3 (Transfer-accounting appendix)**.** *The seed transfers, internal descents, absorbing stops, and finite refinements used in the GEE event graph can be normalized so that no excess atom is counted simultaneously at a source and a target exit, and the total final routing multiplicity for each atom is at most* $\log^{C_{route}+C_{transfer}} X$.

Proof. This is the LaTeX summary of `docs/rh-transfer-accounting-formal-appendix.md`. The allowed operations are finite refinement, source-deleting relabelling, internal descent, and absorption. Relabelling preserves mass and deletes the source copy; internal descent is not final load; absorption stops; refinement has fixed logarithmic overlap. The A, FCT, SC descents are bounded by their integer potentials. Summing the logarithmic refinement losses gives $C_{transfer}$. □

Lemma 8.4 (GEE event no-cycle). *Assume transfer accounting, Baseline-Subtraction, the three internal potentials for A, FCT, SC, the PI/DSO capacity ledger, the $LV/LSMP/CE$ thresholds, and the NRC routing alternatives. Then the GEE event graph has no infinite directed cycle carrying positive final load.*

Proof. An A self-return either decreases the ACC potential or is transferred; an FCT self-return either decreases its Noether potential or is transferred; an SC self-return either decreases its short-cluster potential or is transferred. A PI/DSO self-return consumes excess capacity after subtracting zero frequency. The $CE/LSMP$ nodes do not retain transferred seeds, while LV and NRC are absorbing or routing nodes. For mixed cycles use the lexicographic Lyapunov vector

$$\mathcal{L} = (\mathcal{A}_{pot}, \mathcal{N}, \mathcal{V}, \mathrm{ExcessCap}, \mathrm{ComplexityRank}).$$

Every non-absorbing traversal either decreases one component of $\mathcal{L}$ or removes the atom from the current exit ledger. Since the components are bounded below and the template complexity is finite, an infinite positive-load cycle is impossible. □

Proposition 8.5 (Low-black-box local exits). *Under the unified threshold convention and the transfer-accounting proposition, the exits $LV, CE, LSMP, FCT, SC, A$ do not carry final load except for $O(\Delta \log^{-B_{final}} X)$ absorbing pieces or source-deleted transfers to named analytic exits.*

Proof. This summarizes `docs/rh-local-exit-proofs-formal-appendix.md`. The LV exit is a relative low-volume threshold plus trivial volume estimate. The $CE/LSMP$ exits retain only small-mass, coarea, thin-layer, or square-summable absorbing pieces; all seeds transfer with source deletion. The FCT, SC, A exits are internal descent mechanisms controlled respectively by Noether, short-cluster, and ACC synchronization potentials; non-descent or repeated-template cases transfer to named exits. Thus these six exits are closed at the bookkeeping/threshold level. This proposition does not close the analytic exits PI, DSO, NRC. □

Lemma 8.6 (GEE threshold compatibility). *Let*

$$\begin{aligned} C_{tot} = {} & C_{route} + C_{seed} + C_{dyadic} + C_{Vaaler} + C_{Fourier} + C_{smooth} \\ & + C_{frame} + C_{template} + C_{capacity} + C_{NRC} + C_{coarea} + C_{boundary} + 100. \end{aligned}$$

Choose $B_{final} = 10C_{tot} + 100$ and require every absorbing exit threshold $B_E \geq B_{final} + C_{tot}$. Then all absorbing contributions sum to $O(\Delta \log^{-B_{final}} X)$, in particular $o(\Delta \log^{-C_{route}} X)$ once $B_{final} > C_{route}$, and internal transfers or seed transfers do not change this estimate.

Proof. Each local absorbing estimate is bounded by $\Delta / \log^{B_E - C_E} X$ with $C_E \leq C_{tot}$. All further routing, dyadic, Fourier/Vaaler, boundary, frame, capacity, and seed losses are bounded by another factor $\log^{C_{tot}} X$. Thus the total absorbing contribution is at most $\Delta / \log^{B_{final}} X = o(\Delta \log^{-C_{route}} X)$. Internal descent steps are not final load, and seed transfers only relabel the same excess atom with finite overlap already included in C_{tot}. □

Proposition 8.7 (PI-Lac input). *For strongly lacunary fixed-template projection packs, Mellin support has finite overlap; after zero-frequency baseline subtraction, the lacunary part contributes no positive final PI load except through named terminal failures.*

Proof. This is the LaTeX summary of `docs/rh-pi-lac-input-final.md`. In logarithmic coordinates the windows lie in intervals $[\log X_j - C, \log X_j + C]$. Strong lacunarity makes these intervals finitely overlapping. Fixed CRT/Bohr/reciprocal templates refine inside each interval and do not create cross-scale support. Hence the lacunary energy is bounded by its disjoint zero-frequency capacity; after Baseline-Subtraction this capacity is not final load. Failure of finite overlap or single-scale boundedness is a named $CE/LSMP/LV/SC$ or single-scale PI terminal. □

Proposition 8.8 (AEX-1 projection capacity). *The PI projection-capacity inequality reduces to two capacity inputs: a lacunary Bessel/Parseval bound and a dense fixed-template Carleson/square-function bound, both after zero-frequency baseline subtraction.*

Proof. This summarizes `docs/rh-aex1-projection-capacity-formal.md`. Baseline subtraction changes the PI load into the positive part of $Energy - C_0 Cap^0$. The projection family is split into lacunary and dense scale packs. The lacunary pack is controlled by finite-overlap Bessel/Parseval capacity, while the dense pack is controlled by the fixed-template Carleson/square-function capacity. Applicability failures transfer to named exits by the NoReturn and transfer-accounting ledgers. Thus AEX-1 is reduced to the formal inputs PI-Lac and PI-Dense. □

Proposition 8.9 (PI-Dense reduction). *The dense fixed-template projection capacity input is not an independent final input: it reduces to the CRT martingale square-function ledger DSO-C/TC, the Euler local factor matching DSO-E/EXT-KL, and the recorded DSO-SF baseline capacity input.*

Proof. This summarizes `docs/rh-pi-dense-input-final.md`. Fixed-template dense packs pull back to the CRT inverse-limit space. DSO-C gives martingale orthogonality, while TC verifies natural refinement or transfers to named exits. Euler local complexity is handled by DSO-E: non-resonant pieces use EXT-KL/NRC, resonant failures enter FCT, and small-mass pieces enter LSMP. The quantitative dense capacity input is the recorded DSO-SF theorem, which bounds the martingale square-function baseline after zero-frequency subtraction. □

Proposition 8.10 (AEX-2 DSO square-function). *The DSO square-function inequality is supplied by a pure Parseval/frame ledger plus the recorded DSO-SF capacity theorem controlling the martingale square-function baseline.*

Proof. This summarizes `docs/rh-aex2-dso-squarefunction-formal.md`. Fixed-complexity frequency packages are finite intersections of allowed CRT/Bohr labels, so Parseval controls their squared projections by the relevant martingale differences, up to logarithmic frame losses. Errors that are not square-summable, non-allowed packages, or high-overlap packages transfer to named exits. The quantitative input is DSO-SF: the martingale square-function baseline is bounded by the allowed capacity plus $\Delta^2/\log^B X$. □

Proposition 8.11 (Analytic exit reduction). *After the six low-black-box exits have been closed, the remaining analytic exits PI, DSO, NRC are reduced to three inequalities: a zero-frequency-subtracted projection-capacity bound, a no-power-loss DSO square-function bound, and an NRC entrywise parameter-matching bound.*

Proof. By the baseline-subtraction convention in Definition 8.1, the final PI and DSO loads contain only positive excess over the CRT zero-frequency capacity; ordinary zero-frequency mass has already been removed in PC2 and cannot be registered again. Therefore the analytic load decomposes as

$$\mathrm{Load}(PI;X) + \mathrm{Load}(DSO;X) + \mathrm{Load}(NRC;X) \leq Excess_{PI} + Excess_{DSO} + Load_{NRC} + Err_{tr},$$

where Err_{tr} consists of atoms transferred by the NoReturn and transfer-accounting ledgers to $LV, CE, LSMP, FCT, SC, A$.

AEX-1 gives $Excess_{PI} = O(\Delta \log^{-B_{PI}} X)$ after the lacunary/dense projection split. AEX-2 gives $Excess_{DSO} = O(\Delta \log^{-B_{DSO}} X)$ from the Parseval/frame square-function reduction and the DSO-SF capacity input. AEX-3 treats each NRC entry separately: single-variable entries are square-root-small by the stated EXT estimate, while tail, two-variable, and DSO-E entries either satisfy the same bound after parameter matching or are transferred to the named exits. Transferred atoms are deleted from the source exit before final load is computed, so they are not counted twice. Hence the three analytic exits have total load $o(\Delta \log^{-C_{route}} X)$ after the threshold hierarchy is imposed, by the preceding AEX propositions together with the recorded PI-Lac, PI-Dense, DSO-SF, EXT-KL, transfer-accounting, and restricted external inputs listed in §9. □

Proposition 8.12 (DSO-SF input). *After fixed-template natural refinement and source-deleting transfer of all error/complexity failures, the DSO square-function baseline is the Hilbert martingale identity*

$$\sum_k \|D_k F\|_2^2 \leq \|F - E_0 F\|_2^2.$$

Thus DSO-SF is a formal L^2 martingale capacity theorem, not an additional analytic black box.

Proof. This summarizes `docs/rh-dso-sf-input-final.md`. Conditional expectations onto the CRT filtration are orthogonal projections, so martingale differences are pairwise orthogonal. Fixed-complexity DSO frequency packages are orthogonal subprojections of these differences, up to logarithmic frame overlap. Non-square-summable template errors, non-allowed packets, or excessive overlap transfer to named exits. Hence the remaining DSO square-function mass is bounded by the zero-frequency-subtracted L^2 capacity. □

Proposition 8.13 (AEX-3 NRC parameter matching). *The NRC entrywise parameter condition reduces to four cases: single-variable PPI is square-root-small, Tail/RKS transfers to LV or the main layer, two-variable PPI transfers through MidCap, and DSO-E depends on DSO-SF.*

Proof. This summarizes `docs/rh-aex3-nrc-parameter-match-formal.md`. For single-variable PPI, $P \leq X$ and polylogarithmic complexity give $X^{1/2} \log^C X = o(\Delta)$ because $\beta > 1/2$. Tail/RKS is either low volume or returns to the main layer. Two-variable PPI is not closed by a dangerous $QP^{1/2}$ summation; instead PPI-Rank, Capacity-Match, and MidCap route it to named exits. DSO-E is square-root-small only once DSO-SF is available. □

Theorem 8.14 (GEE upper bound). *Assume the local exit propositions, AEX reductions, EXT-Precision theorem, Lemmas 8.4 and 8.6, and Definition 8.1. Choose the threshold hierarchy so that the final absorption exponent B_{final} satisfies $B_{final} > C_{route} + 10$. Then*

$$\sum_{E \in \mathcal{E}} \mathrm{Load}(E; X) \ll \Delta \log^{-B_{final}} X = o(\Delta \log^{-C_{route}} X).$$

Proof. The absorbing pieces are not merely $o(\Delta)$: by Lemma 8.6 and the threshold hierarchy, every absorbing entry is bounded by $\Delta \log^{-B_{entry}} X$ before global summation. All dyadic, Vaaler, frame, template, seed-transfer, and route-overlap losses are dominated by the fixed constant C_{total}, while B_{final} is chosen larger than $C_{route} + C_{total} + 10$. Hence the sum of all absorbing pieces is $\ll \Delta \log^{-B_{final}} X$. The PI/DSO zero-frequency capacity is not counted as load; only excess capacity remains, and it is either absorbed within the same threshold budget or transferred. The A, FCT, SC pieces terminate by their potentials or transfer to another named exit. Lemma 8.4 excludes positive-load free circulation among these transfers. Thus after finitely many routed descents only absorbing pieces remain, with total $o(\Delta \log^{-C_{route}} X)$. □

Corollary 8.15 (GEE conditional contradiction). *If Lemma 8.2 and Theorem 8.14 use the same load convention, then an off-critical zero has no GEE escape channel.*

Proof. Lemma 8.2 gives a lower bound $\Delta \log^{-C_{route}} X - o(\Delta \log^{-C_{route}} X)$ for the total final load, while Theorem 8.14 gives $o(\Delta \log^{-C_{route}} X)$. These two estimates are incompatible along the successful scales. □

9 External references

The only permitted external inputs are the restricted forms below. Detailed adaptation: `docs/rh-ext-precision-final.md`; citation locations: `docs/rh-u3-ext-reference-table.md`.

Label	Source	Restricted use
EXT-PC1-LI	Titchmarsh–Heath-Brown [Tit86]; Ingham [Ing32]	infinite oscillating subsequence
EXT-KL	Iwaniec–Kowalski Ch. 12 [IK04]; Katz [Kat88]	prime-modulus $ax + b/x$ Weil/Kloosterman and completion
EXT-Vaaler	Vaaler [Vaa85]	one-dimensional interval trigonometric approximation
EXT-BG	Bourgain–Garaev [BG12]; Baker [Bak12]	appendix reciprocal-sum savings
EXT-Selberg	Halberstam–Richert [HR74]; Iwaniec–Kowalski [IK04]	Selberg upper sieve only
EXT-Vaughan	Vaughan [Vau77]; Iwaniec–Kowalski [IK04]	Vaughan identity / Type I–II decomposition

Proposition 9.1 (EXT-Precision theorem). *All external analytic inputs used in this manuscript are confined to the six labelled packages in Section 9: PC1 explicit formula/Landau–Ingham oscillation, prime-modulus Kloosterman–Weil completion, one-dimensional Vaaler approximation, Bourgain–Garaev/Baker reciprocal-sum savings, Selberg upper sieve, and Vaughan Type I/II decomposition.*

Proof. This summarizes `docs/rh-ext-precision-final.md`, with the PC1 oscillation interface further fixed in `docs/rh-ext-pc1-li-precision-final.md`. Each package is used only in the restricted form listed in the table: smooth PC1 oscillation, $ax + b/x$ modulo prime estimates, one-dimensional interval truncation, fixed logarithmic reciprocal-sum savings, upper-bound sieve, and fixed-loss Vaughan decomposition. Degenerate or inapplicable cases are not left as analytic failures; they transfer to the named exits already present in the event graph. The chapter/article-level citation locations are recorded in `docs/rh-u3-ext-reference-table.md`; any remaining monograph page-number check is a submission-format verification task, not a separate mathematical hypothesis. □

10 Consolidated contradiction theorem

Theorem 10.1 (Consolidated contradiction-field theorem). *By Theorems 2.1, 4.5, 5.5, 6.4, 7.6, 8.14, Lemmas 3.1, 3.2, 8.2, and the restricted external inputs in §9, an off-critical zero $\beta > 1/2$ has no free escape channel in the contradiction-field event graph.*

Proof. PC1 gives a smooth Chebyshev anomaly of size $\Delta = X^{\beta-o(1)}$. PC2 transfers it to the rough-composite ledger with opposite sign. Lemma 3.2 splits the covering field into sparse and dense structural branches. The sparse branch is exhausted by Theorem 4.5; the dense branch is exhausted by Theorem 5.5; the internal terminals cannot cycle by Theorem 6.4; Fourier/Vaaler tails do not create a new terminal by Theorem 7.6. The GEE ledger then supplies the global lower bound Lemma 8.2 and the global upper bound Theorem 8.14, so every named exit is either absorbing, descending, or transferred without free circulation. Thus every path enters a controlled exit and the total final load is simultaneously $\geq \Delta \log^{-C_{route}} X - o(\Delta \log^{-C_{route}} X)$ and $o(\Delta \log^{-C_{route}} X)$ under the verified hypotheses. $\square$

Remark 10.2 (Submission warning)**.** This theorem is now stated as a synthesis of the preceding numbered results rather than as a separate assumption. To promote the manuscript to a final unconditional RH theorem, the reviewer must still verify that every cited local theorem, controlled exit, threshold hierarchy, and restricted external input has a fully accepted proof under the stated hypotheses and that no controlled exit hides an unproved positive-power loss. This warning prevents the verification manuscript from overstating the current mathematical status.

References

[Bak12] Roger C. Baker. Kloosterman sums with prime variable. *Acta Arithmetica*, 156(4):351–372, 2012.

[BG12] Jean Bourgain and M. Z. Garaev. Sumsets of reciprocals in prime fields and multilinear kloosterman sums, 2012.

[HR74] Heini Halberstam and Hans-Egon Richert. *Sieve Methods*. Number 4 in London Mathematical Society Monographs. Academic Press, 1974.

[IK04] Henryk Iwaniec and Emmanuel Kowalski. *Analytic Number Theory*, volume 53 of *American Mathematical Society Colloquium Publications*. American Mathematical Society, Providence, RI, 2004.

[Ing32] A. E. Ingham. *The Distribution of Prime Numbers*. Number 30 in Cambridge Tracts in Mathematics and Mathematical Physics. Cambridge University Press, 1932.

[Kat88] Nicholas M. Katz. *Gauss Sums, Kloosterman Sums, and Monodromy Groups*. Annals of Mathematics Studies. Princeton University Press, 1988.

[Tit86] E. C. Titchmarsh. *The Theory of the Riemann Zeta-function*. Oxford University Press, 2 edition, 1986. Revised by D. R. Heath-Brown.

[Vaa85] Jeffrey D. Vaaler. Some extremal functions in fourier analysis. *Bulletin of the American Mathematical Society*, 12(2):183–216, 1985.

[Vau77] R. C. Vaughan. Sommes trigonométriques sur les nombres premiers. *C. R. Acad. Sci. Paris Sér. A-B*, 285:A981–A983, 1977.